

ISHAAN RANJAN

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Builder and researcher creating applied AI systems for messy real-world decision problems, combining machine learning, statistics, forecasting, and usable software. Co-founder of an applied-AI studio with 5,000+ users and \$48K+ in non-dilutive funding, with published research spanning policy forecasting, agent systems, and environmental engineering.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. Computer Science, Threads: Intelligence and Theory | Expected Dec 2028 | Begins Fall 2026

- Accelerated graduation track via AP and transfer credit.

BASIS Peoria

Peoria, AZ

High School Diploma | May 2026

- GPA 3.99 UW / 4.87 W. SAT 1540. ACT 35. 15 AP exams (14 scores of 5, one 4), AP Scholar with Distinction. BASIS Educational Ventures Scholar (\$5,000).

SKILLS

Programming: Python, Java, JavaScript, TypeScript, Swift, SQL, R, HTML/CSS.

AI / ML: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LoRA and supervised fine-tuning, small language models, multi-agent systems, LLM evaluation, prompt engineering, Model Context Protocol (MCP), graph neural networks.

Data / Research: pandas, NumPy, Jupyter, causal inference, difference-in-differences, regression, experimental design, reproducible analysis, data visualization, dashboards.

Product / Infra: FastAPI, React, Electron, Streamlit, Supabase, Vercel, REST APIs, Playwright, Git/GitHub, Arduino/IoT.

Business / Communication: structured problem solving, market and user research, stakeholder interviews, product strategy, KPI dashboards, competitive analysis, technical writing, public speaking.

PROJECTS

PRISM: Alcohol Policy Impact Atlas | Python, Causal Inference, NLP, Streamlit

- Built a 51-state, 2000-2022 policy panel with event-study difference-in-differences models, automating 15+ figures and a public decision-support dashboard.
- Designed classification, causal-audit, and crash-risk forecasting modules with transparent assumptions and reproducible artifacts. [Dashboard](#) | [SSRN](#)

Specialized Small-Model Subagents (Research) | PyTorch, LoRA, Fine-Tuning, MCP

- Fine-tuned small language models into task-specific coding subagents; a 3B specialist matched a 72B baseline at far lower cost per successful task.
- Built 135M-1.5B specialists and an MCP delegation server for low-cost multi-agent coding; paper in progress for an ML/agents workshop.

AgentTree | JavaScript, Electron, AI Agents, Developer Tools

- Built an open-source, local-first workspace for hierarchical master/subagent coding sessions with shared context, live agent status, runtime model switching, and session control; 5,000+ users. [GitHub](#)

YConstruction | SwiftUI, Gemma, Whisper, Supabase, IFC/BIM

- Built a voice-first iPhone app for construction defect reporting: on-device transcription, AI issue structuring, offline queueing, Supabase sync, and BIM-grounded architect review. Selected for a live pitch to YC investors at the YC x Google DeepMind hackathon. [Devpost](#)

DebatICA | NLP, Web Crawling, APIs

- Shipped a debate research and round-prep platform used by 2,000+ debaters; built crawling, deduplication, and source-organization pipelines that cut research time while keeping argument synthesis with the user. [Site](#)

S.A.B.E.R. Bioreactor | Arduino/IoT, Random Forest, Environmental Engineering

- Built an AI-automated *Chlorella vulgaris* bioreactor using pH, turbidity, and CO₂/O₂ sensors with pump-control logic to stabilize growth under variable conditions. AZSEF Grand Award; published preprint. [Paper](#)

PharmaFlow | Graph Neural Networks, Temporal Modeling

- Modeled pharmaceutical payments and prescribing as a temporal graph to flag influence patterns; prototype reached 82% accuracy on a dataset constructed to mirror CMS Open Payments distributions.

EXPERIENCE

Luxen LLC

Dec 2023 - Present

Co-Founder | Applied-AI studio for automation, research, and developer tooling

- Led product strategy from prototype to public deployment across AI agents, developer tools, and research workflows; turned 100+ user interviews into roadmap decisions.
- Raised \$48K+ in non-dilutive grants, startup support, and cloud credits through 1517 Fund, Microsoft for Startups, and BizWorld; earned 1517 Medici recognition and 2x Z Fellows Finalist distinction.
- Scaled products to 5,000+ active users and 1.5M+ media views.

University of Arizona, Precision Aging Network Lab

Summer 2025

KEYS Research Intern | Biostatistics and epigenetic aging

- Analyzed vitamin D, omega-3, and exercise interventions against epigenetic aging clocks (Horvath, AltumAge, DunedinPACE) in R and Python with reproducible notebooks; reported and stress-tested a null result with attention to confounding.
- Presented a poster talk to 800+ attendees, translating biological-aging models into clear visuals for a mixed audience.

LEADERSHIP AND COMMUNITY

- **Speech and Debate Captain:** led a 110+ member team; Arizona State Champion; NSDA Big Questions Top 5 nationally; built case files, drills, and evidence workflows for team-wide performance.
- **Research Club President and Science Olympiad Engineering Lead:** founded a student research program, mentored 30+ students through 10+ papers; Wind Power first place at the Gopher Invitational.
- **AI Literacy Initiative:** co-designed classroom tools and guides with teachers, reaching 150+ students and faculty, focused on reducing teacher workload while preserving student reasoning.
- **MIT Blueprint and University of Austin Palantir Conference:** selected attendee; built GreenhouseAgent (A.C.R.E.), an Arduino/FastAPI greenhouse automation stack, at MIT Blueprint.

PUBLICATIONS

- **Google Scholar:** scholar.google.com/citations?user=pV4cNIQAAAAJ&hl=en
- PRISM: Alcohol Policy Impact Atlas. [SSRN](#)
- Criminal Tribes Act and Social Exclusion (ranked #1 in downloads for relevant eJournals; National History Day national qualifier). [SSRN](#)
- S.A.B.E.R.: Sustainable AI-automated Bioreactor for Environmental Remediation. [Preprint](#)

AWARDS AND HONORS

- **Entrepreneurship:** 1517 Fund Medici Grantee; 2x Z Fellows Finalist (2024, 2025); BizWorld YES! Accelerator; Microsoft for Startups credits; Y Combinator Startup School 2026; \$48K+ in funding and credits.
- **Research and STEM:** Georgia Tech Stamps and Gold Scholarship Semifinalist; BASIS Educational Ventures Scholar (\$5,000); AZSEF Grand Award; SSRN paper ranked #1 in downloads; Science Olympiad Wind Power #1 (Gopher Invitational); National Economics Olympiad Top 43; US Medicine Olympiad Honorable Mention.
- **Speech and Debate:** Arizona State Champion; NSDA Big Questions Top 5 nationally; NSDA Nationals qualifier.
- **Academics:** SAT 1540; ACT 35; 15 AP exams (14 scores of 5, one 4); AP Scholar with Distinction; GPA 3.99 UW / 4.87 W.